

Calculus III — Full Solutions

Chapter 1: Differentiation | Chapter 2: Multiple Integrals

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1. Vectors and Geometry of Space

Exercise 1.1. Find the vector $\overrightarrow{AB}$ and its magnitude for (a) $A(1, -2, 4)$, $B(3, 0, -1)$; (b) $A(-3, 5, 0)$, $B(2, -1, 6)$.

$$(a) \overrightarrow{AB} = \langle 3 - 1, 0 - (-2), -1 - 4 \rangle = \langle 2, 2, -5 \rangle.$$

$$|\overrightarrow{AB}| = \sqrt{4 + 4 + 25} = \sqrt{33}.$$

$$(b) \overrightarrow{AB} = \langle 2 - (-3), -1 - 5, 6 - 0 \rangle = \langle 5, -6, 6 \rangle.$$

$$|\overrightarrow{AB}| = \sqrt{25 + 36 + 36} = \sqrt{97}.$$

Exercise 1.2. Let $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle -1, 4, 2 \rangle$. Find (a) $\mathbf{u} \cdot \mathbf{v}$; (b) angle between them; (c) $\mathbf{u} \times \mathbf{v}$; (d) unit vector in direction of $\mathbf{u}$.

$$(a) \mathbf{u} \cdot \mathbf{v} = 2(-1) + (-1)(4) + 3(2) = -2 - 4 + 6 = \boxed{0}. \text{ The vectors are } \textit{orthogonal}.$$

$$(b) \text{ Since } \mathbf{u} \cdot \mathbf{v} = 0, \text{ the angle is } \theta = \pi/2.$$

(c)

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 3 \\ -1 & 4 & 2 \end{vmatrix} = \mathbf{i}((-1)(2) - (3)(4)) - \mathbf{j}((2)(2) - (3)(-1)) + \mathbf{k}((2)(4) - (-1)(-1)) \\ &= \mathbf{i}(-2 - 12) - \mathbf{j}(4 + 3) + \mathbf{k}(8 - 1) = \langle -14, -7, 7 \rangle. \end{aligned}$$

$$\text{Verification: } \mathbf{u} \cdot \langle -14, -7, 7 \rangle = 2(-14) + (-1)(-7) + 3(7) = -28 + 7 + 21 = 0 \checkmark$$

$$(d) |\mathbf{u}| = \sqrt{4 + 1 + 9} = \sqrt{14}. \text{ Unit vector} = \frac{\mathbf{u}}{|\mathbf{u}|} = \left\langle \frac{2}{\sqrt{14}}, \frac{-1}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right\rangle.$$

Exercise 1.3. (a) Parametric and symmetric equations of the line through $P(-1, 2, 3)$ parallel to $\mathbf{v} = \langle 3, -1, 5 \rangle$; (b) line through $A(1, 0, -2)$ and $B(4, 3, 1)$; (c) intersection of $x = 1 + 2t$, $y = -t$, $z = 3 + t$ with the plane $2x - y + z = 7$.

$$(a) \text{ Parametric: } x = -1 + 3t, y = 2 - t, z = 3 + 5t. \quad \text{Symmetric: } \frac{x + 1}{3} = \frac{y - 2}{-1} = \frac{z - 3}{5}.$$

$$(b) \text{ Direction } \mathbf{v} = \overrightarrow{AB} = \langle 3, 3, 3 \rangle. \text{ Parametric: } x = 1 + 3t, y = 3t, z = -2 + 3t.$$

(c) Substitute into the plane:

$$2(1 + 2t) - (-t) + (3 + t) = 7 \implies 2 + 4t + t + 3 + t = 7 \implies 6t = 2 \implies t = \frac{1}{3}.$$

$$\text{Point: } x = \frac{5}{3}, y = -\frac{1}{3}, z = \frac{10}{3}.$$

Exercise 1.4. (a) Plane through $P(2, -1, 3)$ with $\mathbf{n} = \langle 1, 5, -2 \rangle$; (b) plane through $P(0, 1, 1)$, $Q(1, 0, 1)$, $R(1, 1, 0)$; (c) Are $2x - 3y + z = 4$ and $4x - 6y + 2z = 1$ parallel?

(a) $(x - 2) + 5(y + 1) - 2(z - 3) = 0 \implies x + 5y - 2z + 9 = 0.$

(b) $\overrightarrow{PQ} = \langle 1, -1, 0 \rangle$, $\overrightarrow{PR} = \langle 1, 0, -1 \rangle.$

$$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} = \langle (-1)(-1) - 0, 0(1) - 1(-1), 0 - (-1) \rangle = \langle 1, 1, 1 \rangle.$$

Plane: $(x - 0) + (y - 1) + (z - 1) = 0 \implies x + y + z = 2.$

(c) Rewrite the second plane: $2x - 3y + z = \frac{1}{2}$. Both have normal $\langle 2, -3, 1 \rangle$ but different constants ($4 \neq \frac{1}{2}$), so they are **parallel** (and distinct).

Exercise 1.5. Describe and sketch: (a) $x^2 + z^2 = 9$; (b) $z = 4 - x^2 - y^2$; (c) $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$.

(a) Circular cylinder of radius 3 with axis along the y -axis.

(b) Circular paraboloid opening downward, vertex at $(0, 0, 4)$.

(c) Ellipsoid with semi-axes $a = 2$ (along x), $b = 3$ (along y), $c = 4$ (along z).

2. Functions of Several Variables

Exercise 2.1. For each function find the domain, range, and level curves: (a) $f(x, y) = \sqrt{16 - x^2 - y^2}$; (b) $f(x, y) = \ln(x + y - 1)$; (c) $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$.

(a) Domain: $x^2 + y^2 \leq 16$ (disk of radius 4). Range: $[0, 4]$. Level curves $\sqrt{16 - x^2 - y^2} = k$: circles $x^2 + y^2 = 16 - k^2$, $0 \leq k \leq 4$.

(b) Domain: $x + y > 1$ (open half-plane). Range: $\mathbb{R}$. Level curves: $x + y = 1 + e^k$ (parallel lines).

(c) Domain: $\mathbb{R}^2 \setminus \{(0, 0)\}$. Writing in polar: $f = \cos(2\theta)$, so Range = $[-1, 1]$. Level curves: $x^2(1 - k) = y^2(1 + k)$, i.e. $y = \pm x\sqrt{\frac{1-k}{1+k}}$ for $|k| < 1$ (lines through origin); $k = 1$ gives $y = 0$; $k = -1$ gives $x = 0$.

Exercise 2.2. Find the domain and level surfaces of $f(x, y, z) = \sqrt{9 - x^2 - y^2 - z^2}$.

Domain: $x^2 + y^2 + z^2 \leq 9$ (closed ball of radius 3). Level surfaces $f = k$: $x^2 + y^2 + z^2 = 9 - k^2$, spheres of radius $\sqrt{9 - k^2}$ for $0 \leq k \leq 3$.

Exercise 2.3. Sketch the level curves of $f(x, y) = x^2 + 4y^2$ for $k = 0, 1, 4, 9$.

Setting $x^2 + 4y^2 = k$:

- $k = 0$: single point $(0, 0)$.
- $k = 1$: ellipse $x^2/1 + y^2/(1/4) = 1$, semi-axes $a = 1$, $b = \frac{1}{2}$.
- $k = 4$: ellipse $x^2/4 + y^2/1 = 1$, semi-axes $a = 2$, $b = 1$.
- $k = 9$: ellipse $x^2/9 + y^2/(9/4) = 1$, semi-axes $a = 3$, $b = \frac{3}{2}$.

The level curves are concentric ellipses expanding outward.

3. Limits and Continuity

Exercise 3.1. Evaluate each limit or show it does not exist.

(a) $\lim_{(x,y) \rightarrow (1,2)} (x^3y - 2xy^2 + 5)$. The function is a polynomial, hence continuous everywhere:

$$= 1^3 \cdot 2 - 2 \cdot 1 \cdot 4 + 5 = 2 - 8 + 5 = -1.$$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{5x^2y}{x^2 + y^2}$. Using polar coordinates $x = r \cos \theta$, $y = r \sin \theta$:

$$\left| \frac{5x^2y}{x^2 + y^2} \right| = \frac{5r^2 \cos^2 \theta \cdot |r \sin \theta|}{r^2} = 5r |\cos^2 \theta \sin \theta| \leq 5r \rightarrow 0.$$

By Squeeze Theorem, the limit is $\boxed{0}$.

(c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$. Factor: $x^2 - xy = x(x - y) = x(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y})$.

Hence

$$\frac{x^2 - xy}{\sqrt{x} - \sqrt{y}} = x(\sqrt{x} + \sqrt{y}) \rightarrow 0 \cdot (0 + 0) = \boxed{0}.$$

(d) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$. Along $y = 0$: limit = 0. Along $x = y^2$: $\frac{y^2 \cdot y^2}{y^4 + y^4} = \frac{y^4}{2y^4} = \frac{1}{2}$. Different limits $\Rightarrow$ limit **does not exist**.

Exercise 3.2. Prove by the Squeeze Theorem that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y^2}{x^2 + y^2} = 0$.

Since $x^2 \leq x^2 + y^2$, we have $x^2y^2 \leq (x^2 + y^2)y^2$. Therefore

$$0 \leq \frac{x^2y^2}{x^2 + y^2} \leq y^2.$$

Since $y^2 \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$, the Squeeze Theorem gives the limit = 0. $\square$

Exercise 3.3. Discuss the continuity of $f(x, y) = \begin{cases} 2xy/(x^2 + y^2) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$.

For $(x, y) \neq (0, 0)$: f is a ratio of polynomials with nonzero denominator, hence continuous.

At $(0, 0)$: along $y = 0$, $\lim f = 0$; along $y = x$, $\lim f = 2x^2/(2x^2) = 1 \neq 0$. The limit does not exist (or does not equal $f(0, 0) = 0$), so f is **discontinuous at** $(0, 0)$.

4. Partial Derivatives and Higher Derivatives

Exercise 4.1. Find all first-order partial derivatives.

(a) $f = x^4y - 3x^2y^3 + 7y$:

$$f_x = 4x^3y - 6xy^3, \quad f_y = x^4 - 9x^2y^2 + 7.$$

(b) $g = e^x \sin(xy)$:

$$g_x = e^x \sin(xy) + e^x \cdot y \cos(xy) = e^x (\sin(xy) + y \cos(xy)),$$

$$g_y = e^x \cdot x \cos(xy) = xe^x \cos(xy).$$

(c) $h = xy^2z^3 + \ln(x + 2y + 3z)$:

$$h_x = y^2z^3 + \frac{1}{x + 2y + 3z}, \quad h_y = 2xyz^3 + \frac{2}{x + 2y + 3z}, \quad h_z = 3xy^2z^2 + \frac{3}{x + 2y + 3z}.$$

(d) $w = s \arctan(t/s)$. Let $u = t/s$; $\partial u / \partial s = -t/s^2$, $\partial u / \partial t = 1/s$.

$$w_s = \arctan(t/s) + s \cdot \frac{1}{1 + t^2/s^2} \cdot \frac{-t}{s^2} = \arctan(t/s) - \frac{t}{s^2 + t^2},$$

$$w_t = s \cdot \frac{1}{1 + t^2/s^2} \cdot \frac{1}{s} = \frac{s^2}{s^2 + t^2}.$$

Exercise 4.2. Find all second-order partial derivatives of $f(x, y) = x^3y^2 - 2x^2y^4 + 5xy$ and verify Clairaut's theorem ($f_{xy} = f_{yx}$).

First-order: $f_x = 3x^2y^2 - 4xy^4 + 5y$, $f_y = 2x^3y - 8x^2y^3 + 5x$.

Second-order:

$$\begin{aligned}f_{xx} &= 6xy^2 - 4y^4, \\f_{yy} &= 2x^3 - 24x^2y^2, \\f_{xy} &= \frac{\partial}{\partial y}(3x^2y^2 - 4xy^4 + 5y) = 6x^2y - 16xy^3 + 5, \\f_{yx} &= \frac{\partial}{\partial x}(2x^3y - 8x^2y^3 + 5x) = 6x^2y - 16xy^3 + 5.\end{aligned}$$

Since $f_{xy} = f_{yx} = 6x^2y - 16xy^3 + 5$, Clairaut's theorem is verified. ✓

Exercise 4.3. Verify that the following functions are harmonic ($f_{xx} + f_{yy} = 0$).

(a) $f = x^3 - 3xy^2$: $f_{xx} = 6x$, $f_{yy} = -6x$. Sum = 0. ✓

(b) $f = e^x \cos y$: $f_{xx} = e^x \cos y$, $f_{yy} = -e^x \cos y$. Sum = 0. ✓

(c) $f = \ln(x^2 + y^2)$:

$$f_x = \frac{2x}{x^2 + y^2}, \quad f_{xx} = \frac{2(x^2 + y^2) - 4x^2}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}.$$

By symmetry, $f_{yy} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$. $f_{xx} + f_{yy} = 0$. ✓

Exercise 4.4. Let $f(x, y) = x^2e^y + y \sin x$. Calculate f_{xxy} and f_{xyy} .

$$f_x = 2xe^y + y \cos x. \quad f_{xx} = 2e^y - y \sin x.$$

$$f_{xxy} = \frac{\partial}{\partial y}(2e^y - y \sin x) = 2e^y - \sin x.$$

$$f_{xy} = \frac{\partial}{\partial y}(2xe^y + y \cos x) = 2xe^y + \cos x.$$

$$f_{xyy} = \frac{\partial}{\partial y}(2xe^y + \cos x) = 2xe^y.$$

5. Tangent Planes and Linear Approximations

Exercise 5.1. (a) Tangent plane to $z = x^2 + 2y^2 - 3x$ at $(1, 1, 0)$; (b) tangent plane to $z = e^{x-y}$ at $(2, 2, 1)$.

(a) $f_x = 2x - 3$, $f_y = 4y$. At $(1, 1)$: $f_x = -1$, $f_y = 4$.

$$z - 0 = -1(x - 1) + 4(y - 1) \implies \boxed{z = -x + 4y - 3}.$$

(b) $f_x = e^{x-y}$, $f_y = -e^{x-y}$. At $(2, 2)$: $f_x = 1$, $f_y = -1$.

$$z - 1 = 1(x - 2) + (-1)(y - 2) \implies \boxed{z = x - y + 1}.$$

Exercise 5.2. Find the linearisation of $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$ and estimate $f(1.95, 1.05)$.

$$f(2, 1) = \sqrt{20 - 4 - 7} = 3.$$

$$f_x = \frac{-x}{\sqrt{20 - x^2 - 7y^2}} \bigg|_{(2,1)} = \frac{-2}{3}, \quad f_y = \frac{-7y}{\sqrt{20 - x^2 - 7y^2}} \bigg|_{(2,1)} = \frac{-7}{3}.$$

$$L(x, y) = 3 - \frac{2}{3}(x - 2) - \frac{7}{3}(y - 1).$$

$$f(1.95, 1.05) \approx L(1.95, 1.05) = 3 - \frac{2}{3}(-0.05) - \frac{7}{3}(0.05) = 3 + \frac{0.10}{3} - \frac{0.35}{3} = 3 - \frac{0.25}{3} \approx 2.917.$$

Exercise 5.3. Box $80 \times 60 \times 50$ cm, error ± 0.2 cm in each dimension. Estimate maximum error in volume.

$$V = lwh, \quad dV = wh \, dl + lh \, dw + lw \, dh. \quad \text{Maximum error:}$$

$$|dV| \leq (wh + lh + lw) \cdot 0.2 = (60 \cdot 50 + 80 \cdot 50 + 80 \cdot 60) \cdot 0.2 = (3000 + 4000 + 4800) \cdot 0.2 = 11800 \cdot 0.2 = \boxed{2360 \text{ cm}^3}.$$

Exercise 5.4. Find the total differential dz for (a) $z = x^4y + x\sqrt{y}$; (b) $z = \arctan\left(\frac{x}{y}\right)$.

$$(a) \quad \frac{\partial z}{\partial x} = 4x^3y + \sqrt{y}, \quad \frac{\partial z}{\partial y} = x^4 + \frac{x}{2\sqrt{y}}.$$

$$dz = (4x^3y + \sqrt{y}) \, dx + \left(x^4 + \frac{x}{2\sqrt{y}}\right) \, dy.$$

$$(b) \quad \frac{\partial z}{\partial x} = \frac{1}{1 + x^2/y^2} \cdot \frac{1}{y} = \frac{y}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{1}{1 + x^2/y^2} \cdot \frac{-x}{y^2} = \frac{-x}{x^2 + y^2}.$$

$$dz = \frac{y}{x^2 + y^2} \, dx - \frac{x}{x^2 + y^2} \, dy.$$

6. The Chain Rule

Exercise 6.1. Use the Chain Rule to find dz/dt .

(a) $z = x^2y + 3xy^4$, $x = \sin t$, $y = e^t$.

$$\frac{dz}{dt} = \underbrace{(2xy + 3y^4)}_{\partial z / \partial x} \cos t + \underbrace{(x^2 + 12xy^3)}_{\partial z / \partial y} e^t.$$

(General formula; substituting $x = \sin t$, $y = e^t$ gives the explicit form.)

(b) $z = \sqrt{x^2 + y^2}$, $x = e^t \cos t$, $y = e^t \sin t$. Note $x^2 + y^2 = e^{2t}$, so $z = e^t$.

$$\frac{dz}{dt} = \frac{x}{e^t}(e^t \cos t - e^t \sin t) + \frac{y}{e^t}(e^t \sin t + e^t \cos t).$$

$$= e^t \cos t(\cos t - \sin t) + e^t \sin t(\sin t + \cos t) = e^t(\cos^2 t + \sin^2 t) = \boxed{e^t}.$$

Exercise 6.2. Use the Chain Rule to find $\partial z / \partial s$ and $\partial z / \partial t$.

(a) $z = e^{2x+3y}$, $x = s/t$, $y = t/s$.

$$\frac{\partial z}{\partial s} = e^{2x+3y} \left(2 \cdot \frac{1}{t} + 3 \cdot \frac{-t}{s^2} \right) = e^{2s/t+3t/s} \left(\frac{2}{t} - \frac{3t}{s^2} \right).$$

$$\frac{\partial z}{\partial t} = e^{2x+3y} \left(2 \cdot \frac{-s}{t^2} + 3 \cdot \frac{1}{s} \right) = e^{2s/t+3t/s} \left(\frac{3}{s} - \frac{2s}{t^2} \right).$$

(b) $z = x^2 \tan y$, $x = st^2$, $y = s^2t$.

$$\frac{\partial z}{\partial s} = 2x \tan y \cdot t^2 + x^2 \sec^2 y \cdot 2st = 2st^4 \tan(s^2t) + 2s^3t^5 \sec^2(s^2t),$$

$$\frac{\partial z}{\partial t} = 2x \tan y \cdot 2st + x^2 \sec^2 y \cdot s^2 = 4s^2t^3 \tan(s^2t) + s^4t^4 \sec^2(s^2t).$$

Exercise 6.3 (Implicit Differentiation). (a) Find dy/dx if $x^4 + x^2y^2 + y^4 = 3$; (b) find $\partial z / \partial x$ and $\partial z / \partial y$ if $x^2 + 2y^2 + 3z^2 + xyz = 6$.

(a) Let $F = x^4 + x^2y^2 + y^4 - 3$. $F_x = 4x^3 + 2xy^2$, $F_y = 2x^2y + 4y^3$.

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{4x^3 + 2xy^2}{2x^2y + 4y^3} = -\frac{2x^3 + xy^2}{x^2y + 2y^3}.$$

(b) Let $F = x^2 + 2y^2 + 3z^2 + xyz - 6$. $F_x = 2x + yz$, $F_y = 4y + xz$, $F_z = 6z + xy$.

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2x + yz}{6z + xy}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{4y + xz}{6z + xy}.$$

7. Directional Derivatives and the Gradient

Exercise 7.1. Find the directional derivative at the given point in the given direction.

(a) $f = x^2y^3 - 2x$, point $(1, -1)$, direction angle $\theta = \pi/4$. Unit vector $\mathbf{u} = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$.
 $f_x = 2xy^3 - 2$, $f_y = 3x^2y^2$. At $(1, -1)$: $f_x = -4$, $f_y = 3$.

$$D_{\mathbf{u}}f = (-4)\frac{1}{\sqrt{2}} + 3\frac{1}{\sqrt{2}} = \frac{-1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}.$$

(b) $f = x^2yz + xz^2$, point $(1, 1, 1)$, $\mathbf{v} = \langle 1, 2, -1 \rangle$. $|\mathbf{v}| = \sqrt{6}$, $\mathbf{u} = \langle 1, 2, -1 \rangle / \sqrt{6}$.
 $f_x = 2xyz + z^2 = 3$, $f_y = x^2z = 1$, $f_z = x^2y + 2xz = 3$ at $(1, 1, 1)$.

$$D_{\mathbf{u}}f = \frac{3(1) + 1(2) + 3(-1)}{\sqrt{6}} = \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}.$$

Exercise 7.2. (a) Gradient of $f = \sin(x + 2y)$ at $(\pi/6, 0)$; (b) maximum rate of change of $f = \sqrt{x^2 + y^2 + z^2}$ at $(1, 2, 2)$.

(a) $\nabla f = \langle \cos(x+2y), 2\cos(x+2y) \rangle$. At $(\pi/6, 0)$: $x+2y = \pi/6$, so $\nabla f = \langle \frac{\sqrt{3}}{2}, \sqrt{3} \rangle$.

(b) $\nabla f = \frac{\langle x, y, z \rangle}{\sqrt{x^2 + y^2 + z^2}}$. At $(1, 2, 2)$: $|\nabla f| = \frac{\sqrt{1+4+4}}{3} = 1$. Maximum rate of change = 1, in direction $\langle 1, 2, 2 \rangle / 3$.

Exercise 7.3. $T(x, y) = 200e^{-(x^2+y^2)/100}$ at $(3, 4)$: (a) direction of fastest warming; (b) rate in that direction; (c) rate in direction $\mathbf{u} = \langle 4, 3 \rangle / 5$.

$$\nabla T = 200e^{-(x^2+y^2)/100} \cdot \left\langle \frac{-2x}{100}, \frac{-2y}{100} \right\rangle = -4e^{-(x^2+y^2)/100} \langle x, y \rangle.$$

At $(3, 4)$: $x^2 + y^2 = 25$, $e^{-1/4}$. $\nabla T(3, 4) = -4e^{-1/4} \langle 3, 4 \rangle$.

(a) Fastest warming: in direction $-\langle 3, 4 \rangle / 5$ (toward origin, since ∇T points inward).

(b) $|\nabla T| = 4e^{-1/4} \cdot 5 = 20e^{-1/4} \approx 15.6$ per unit length.

(c) $D_{\mathbf{u}}T = \nabla T \cdot \mathbf{u} = (-4e^{-1/4}) \langle 3, 4 \rangle \cdot \frac{\langle 4, 3 \rangle}{5} = \frac{-4e^{-1/4}(12+12)}{5} = -\frac{96e^{-1/4}}{5} \approx -14.9$.

8. Extrema of Functions of Several Variables

Exercise 8.1. Find and classify all critical points of each function.

(a) $f = x^2 + xy + y^2 - 3x$. Setting $f_x = 2x + y - 3 = 0$ and $f_y = x + 2y = 0$: from f_y , $x = -2y$; substituting: $-4y + y - 3 = 0 \Rightarrow y = 1$, $x = -2$. $D = f_{xx}f_{yy} - f_{xy}^2 = 2 \cdot 2 - 1 = 3 > 0$, $f_{xx} = 2 > 0$: **local min** at $(-2, 1)$, value $f(-2, 1) = 4 - 2 + 1 + 6 = 9$.
 (b) $f = 4xy - x^4 - y^4$. $f_x = 4y - 4x^3 = 0$, $f_y = 4x - 4y^3 = 0$. From these: $y = x^3$ and $x = y^3$, giving $x = x^9 \Rightarrow x = 0, \pm 1$.

• $(0, 0)$: $D = (-12x^2)(-12y^2) - 16 \Big|_{(0,0)} = -16 < 0$: **saddle point**.

- $(1, 1)$: $D = 144 - 16 = 128 > 0$, $f_{xx} = -12 < 0$: **local max**, $f(1, 1) = 2$.
- $(-1, -1)$: $D = 128 > 0$, $f_{xx} = -12 < 0$: **local max**, $f(-1, -1) = 2$.

(c) $f = e^x(x + y^2 + 2y)$. Since $e^x > 0$: $f_x = e^x(x + 1 + y^2 + 2y) = 0 \Rightarrow x = -(y + 1)^2$.
 $f_y = e^x(2y + 2) = 0 \Rightarrow y = -1$. At $y = -1$: $x = 0$. Critical point: $(0, -1)$.
 $f_{xx} = e^x(x + 2 + y^2 + 2y)$, $f_{yy} = 2e^x$, $f_{xy} = e^x(2y + 2)$. At $(0, -1)$: $f_{xx} = 1$, $f_{yy} = 2$,
 $f_{xy} = 0$. $D = 2 > 0$, $f_{xx} > 0$: **local min** at $(0, -1)$, value $f(0, -1) = -1$.

Exercise 8.2. Absolute max/min of $f(x, y) = x^2 - 2xy + 2y$ on the triangle with vertices $(0, 0)$, $(3, 0)$, $(3, 3)$.

Interior: $f_x = 2x - 2y = 0$, $f_y = -2x + 2 = 0 \Rightarrow x = 1, y = 1$. $f(1, 1) = 1$.

Boundary:

- $y = 0, 0 \leq x \leq 3$: $f = x^2$. Min 0 at $x = 0$; max 9 at $x = 3$.
- $x = 3, 0 \leq y \leq 3$: $f = 9 - 6y + 2y = 9 - 4y$. Max 9 at $y = 0$; min -3 at $y = 3$.
- $y = x, 0 \leq x \leq 3$: $f = -x^2 + 2x$. $f' = -2x + 2 = 0 \Rightarrow x = 1$, $f(1) = 1$.
Endpoints: 0 and -3 .

Absolute max = 9 at $(3, 0)$. **Absolute min** = -3 at $(3, 3)$.

Exercise 8.3. Find three positive numbers a, b, c with $a + b + c = 100$ maximising abc .

By the AM-GM inequality: $\frac{a + b + c}{3} \geq (abc)^{1/3}$, with equality iff $a = b = c$.

Setting $a = b = c = \frac{100}{3}$ gives the maximum product $\left(\frac{100}{3}\right)^3 = \frac{10^6}{27} \approx 37037$.

9. Lagrange Multipliers

Exercise 9.1. Use Lagrange multipliers.

(a) Maximise $f = xy$ subject to $g = x + 2y - 8 = 0$. $\nabla f = \lambda \nabla g$: $(y, x) = \lambda(1, 2)$, so $y = \lambda$ and $x = 2\lambda \Rightarrow x = 2y$. Substituting: $2y + 2y = 8 \Rightarrow y = 2, x = 4$. Maximum $f(4, 2) = 8$.

(b) Minimise $f = x^2 + y^2 + z^2$ subject to $x + y + z = 12$. $(2x, 2y, 2z) = \lambda(1, 1, 1) \Rightarrow x = y = z = \lambda/2$. Sum: $3\lambda/2 = 12 \Rightarrow \lambda = 8, x = y = z = 4$. Minimum $f(4, 4, 4) = 48$.

(c) Extrema of $f = x^2 + 2y^2$ on $x^2 + y^2 = 1$. $(2x, 4y) = \lambda(2x, 2y)$.

- $x \neq 0$: $\lambda = 1$, then $4y = 2y \Rightarrow y = 0, x = \pm 1$: $f = 1$.

- $x = 0$: $y = \pm 1$: $f = 2$.

Minimum = 1 at $(\pm 1, 0)$. **Maximum** = 2 at $(0, \pm 1)$.

Exercise 9.2. Nearest point on the plane $2x + y + z = 5$ to the origin.

Minimise $f = x^2 + y^2 + z^2$ subject to $g = 2x + y + z - 5 = 0$. $(2x, 2y, 2z) = \lambda(2, 1, 1) \Rightarrow x = \lambda, y = \frac{\lambda}{2}, z = \frac{\lambda}{2}$. Substituting: $2\lambda + \frac{\lambda}{2} + \frac{\lambda}{2} = 5 \Rightarrow 3\lambda = 5 \Rightarrow \lambda = \frac{5}{3}$.

Nearest point: $\left(\frac{5}{3}, \frac{5}{6}, \frac{5}{6}\right)$.

Exercise 9.3. Open-top rectangular box from 12 m^2 of cardboard; maximise volume.

Dimensions: x, y, z (height z). Surface: $g = xy + 2xz + 2yz = 12$. Maximise $V = xyz$. $\nabla V = \lambda \nabla g$:

$$yz = \lambda(y + 2z), \quad xz = \lambda(x + 2z), \quad xy = \lambda(2x + 2y).$$

From the first two: $yz(x + 2z) = xz(y + 2z) \Rightarrow xyz + 2yz^2 = xyz + 2xz^2 \Rightarrow y = x$. Substituting $y = x$: $xz = \lambda(x + 2z)$ and $x^2 = 4\lambda x \Rightarrow \lambda = x/4$. Then $xz = (x/4)(x + 2z) \Rightarrow 4xz = x^2 + 2xz \Rightarrow 2xz = x^2 \Rightarrow z = x/2$. Surface: $x^2 + 2x(x/2) + 2x(x/2) = 12 \Rightarrow 3x^2 = 12 \Rightarrow x = 2$.

Optimal dimensions: $x = y = 2 \text{ m}$, $z = 1 \text{ m}$; $V_{\max} = 4 \text{ m}^3$.

10. Double Integrals over Rectangles

Exercise 10.1. Evaluate the iterated integrals.

(a) $\int_0^3 \int_1^2 (x^2y - 2xy^3) \, dy \, dx.$

$$\int_1^2 (x^2y - 2xy^3) \, dy = \left[\frac{x^2y^2}{2} - \frac{xy^4}{2} \right]_1^2 = 2x^2 - 8x - \frac{x^2}{2} + \frac{x}{2} = \frac{3x^2}{2} - \frac{15x}{2}.$$

$$\int_0^3 \left(\frac{3x^2}{2} - \frac{15x}{2} \right) \, dx = \left[\frac{x^3}{2} - \frac{15x^2}{4} \right]_0^3 = \frac{27}{2} - \frac{135}{4} = -\frac{81}{4}.$$

(b) $\int_0^\pi \int_0^1 y \sin(xy) \, dy \, dx.$

$$\int_0^1 y \sin(xy) \, dy = [-\cos(xy)]_0^1 = 1 - \cos x.$$

$$\int_0^\pi (1 - \cos x) \, dx = [x - \sin x]_0^\pi = \pi.$$

(c) $\int_0^1 \int_0^1 x e^{xy} dy dx.$

$$\int_0^1 x e^{xy} dy = \left[e^{xy} \right]_0^1 = e^x - 1.$$

$$\int_0^1 (e^x - 1) dx = \left[e^x - x \right]_0^1 = e - 1 - 1 = e - 2.$$

Exercise 10.2. Evaluate $\iint_R (x + 2y) dA$, $R = [0, 2] \times [0, 3]$ in both orders.

Order $dy dx$: $\int_0^2 \int_0^3 (x + 2y) dy dx = \int_0^2 \left[xy + y^2 \right]_0^3 dx = \int_0^2 (3x + 9) dx = \left[\frac{3x^2}{2} + 9x \right]_0^2 = 6 + 18 = 24.$

Order $dx dy$: $\int_0^3 \int_0^2 (x + 2y) dx dy = \int_0^3 \left[\frac{x^2}{2} + 2xy \right]_0^2 dy = \int_0^3 (2 + 4y) dy = \left[2y + 2y^2 \right]_0^3 = 6 + 18 = 24.$

Both orders give $\boxed{24}$. ✓

Exercise 10.3. Volume above $R = [-1, 1] \times [0, 2]$ and below $z = 2 + x + 3y$.

$$\begin{aligned} V &= \int_{-1}^1 \int_0^2 (2 + x + 3y) dy dx = \int_{-1}^1 \left[2y + xy + \frac{3y^2}{2} \right]_0^2 dx = \int_{-1}^1 (4 + 2x + 6) dx = \int_{-1}^1 (10 + 2x) dx \\ &= \left[10x + x^2 \right]_{-1}^1 = (10 + 1) - (-10 + 1) = \boxed{20}. \end{aligned}$$

11. Double Integrals over General Regions

Exercise 11.1. Evaluate $\iint_D f dA$ over each described region.

(a) D : bounded by $y = x^2$ and $y = x + 2$. Intersections: $x = -1, 2$.

$$\begin{aligned} \int_{-1}^2 \int_{x^2}^{x+2} (2x - y) dy dx &= \int_{-1}^2 \left[2xy - \frac{y^2}{2} \right]_{x^2}^{x+2} dx \\ &= \int_{-1}^2 \left(2x(x+2) - \frac{(x+2)^2}{2} - 2x^3 + \frac{x^4}{2} \right) dx = \int_{-1}^2 \left(\frac{x^4}{2} - 2x^3 + \frac{3x^2}{2} + 2x - 2 \right) dx \\ &= \left[\frac{x^5}{10} - \frac{x^4}{2} + \frac{x^3}{2} + x^2 - 2x \right]_{-1}^2 = \left(\frac{32}{10} - 8 + 4 + 4 - 4 \right) - \left(\frac{-1}{10} - \frac{1}{2} - \frac{1}{2} + 1 + 2 \right) \\ &= \frac{32}{10} - 4 - \left(\frac{-1}{10} + 2 \right) = \frac{33}{10} - 6 = -\frac{27}{10}. \end{aligned}$$

(b) D : bounded by $y = x$ and $y = x^2$, $0 \leq x \leq 1$.

$$\int_0^1 \int_{x^2}^x xy dy dx = \int_0^1 x \cdot \frac{x^2 - x^4}{2} dx = \frac{1}{2} \int_0^1 (x^3 - x^5) dx = \frac{1}{2} \left[\frac{1}{4} - \frac{1}{6} \right] = \frac{1}{24}.$$

(c) D : triangle $(0, 0), (1, 0), (0, 1)$; $0 \leq x \leq 1, 0 \leq y \leq 1 - x$.

$$\int_0^1 \int_0^{1-x} x \, dy \, dx = \int_0^1 x(1-x) \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{6}.$$

Exercise 11.2. Reverse the order and evaluate.

(a) $\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx$. Region: $0 \leq y \leq 1, 0 \leq x \leq y$.

$$= \int_0^1 \int_0^y \sin(y^2) \, dx \, dy = \int_0^1 y \sin(y^2) \, dy = \left[-\frac{\cos(y^2)}{2} \right]_0^1 = \frac{1 - \cos 1}{2}.$$

(b) $\int_0^4 \int_{\sqrt{y}}^2 e^{x^3} \, dx \, dy$. Region: $0 \leq x \leq 2, 0 \leq y \leq x^2$.

$$= \int_0^2 \int_0^{x^2} e^{x^3} \, dy \, dx = \int_0^2 x^2 e^{x^3} \, dx = \left[\frac{e^{x^3}}{3} \right]_0^2 = \frac{e^8 - 1}{3}.$$

Exercise 11.3. Area between $y = 3x - x^2$ and $y = x$.

Intersections: $3x - x^2 = x \Rightarrow 2x - x^2 = 0 \Rightarrow x = 0, 2$.

$$\text{Area} = \int_0^2 [(3x - x^2) - x] \, dx = \int_0^2 (2x - x^2) \, dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$$

12. Double Integrals in Polar Coordinates

Exercise 12.1. Convert to polar and evaluate.

(a) $\iint_D \sqrt{x^2 + y^2} \, dA$, D : right half-disk of radius 2. $r \in [0, 2], \theta \in [-\pi/2, \pi/2]$.

$$= \int_{-\pi/2}^{\pi/2} \int_0^2 r \cdot r \, dr \, d\theta = \pi \cdot \left[\frac{r^3}{3} \right]_0^2 = \frac{8\pi}{3}.$$

(b) $\iint_D e^{-(x^2+y^2)} \, dA$, D : disk of radius a .

$$= \int_0^{2\pi} \int_0^a e^{-r^2} r \, dr \, d\theta = 2\pi \left[-\frac{e^{-r^2}}{2} \right]_0^a = \pi(1 - e^{-a^2}).$$

(c) $\int_0^2 \int_0^{\sqrt{4-x^2}} (x^2 + y^2)^{3/2} \, dy \, dx$. Upper semicircle: $r \in [0, 2], \theta \in [0, \pi]$.

$$= \int_0^\pi \int_0^2 r^3 \cdot r \, dr \, d\theta = \pi \left[\frac{r^5}{5} \right]_0^2 = \frac{32\pi}{5}.$$

Exercise 12.2. Volume under $z = 9 - x^2 - y^2$, above xy -plane, inside $x^2 + y^2 = 4$.

$$V = \int_0^{2\pi} \int_0^2 (9-r^2)r \, dr \, d\theta = 2\pi \int_0^2 (9r-r^3) \, dr = 2\pi \left[\frac{9r^2}{2} - \frac{r^4}{4} \right]_0^2 = 2\pi(18-4) = \boxed{28\pi}.$$

Exercise 12.3. Area inside $r = 1 + \cos \theta$ and outside $r = 1$.

Intersection: $\cos \theta = 0 \Rightarrow \theta = \pm\pi/2$.

$$\begin{aligned} A &= \frac{1}{2} \int_{-\pi/2}^{\pi/2} [(1 + \cos \theta)^2 - 1] \, d\theta = \frac{1}{2} \int_{-\pi/2}^{\pi/2} (2 \cos \theta + \cos^2 \theta) \, d\theta. \\ &= \frac{1}{2} \left[2 \sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{-\pi/2}^{\pi/2} = \frac{1}{2} \left[\left(2 + \frac{\pi}{4} \right) - \left(-2 - \frac{\pi}{4} \right) \right] = \frac{1}{2} \left(4 + \frac{\pi}{2} \right) = 2 + \frac{\pi}{4}. \end{aligned}$$

13. Triple Integrals

Exercise 13.1. Evaluate the triple integrals.

(a) $\int_0^2 \int_0^x \int_0^{x+y} 6xy \, dz \, dy \, dx$.

$$\int_0^{x+y} 6xy \, dz = 6xy(x+y).$$

$$\int_0^x 6xy(x+y) \, dy = 6x \int_0^x (xy + y^2) \, dy = 6x \left[\frac{xy^2}{2} + \frac{y^3}{3} \right]_0^x = 6x \left(\frac{x^3}{2} + \frac{x^3}{3} \right) = 5x^5.$$

$$\int_0^2 5x^5 \, dx = \left[x^5 \right]_0^2 = 32.$$

(b) $\iiint_B x e^{y+2z} \, dV$, $B = [0, 1]^3$. Separable:

$$= \left(\int_0^1 x \, dx \right) \left(\int_0^1 e^y \, dy \right) \left(\int_0^1 e^{2z} \, dz \right) = \frac{1}{2} \cdot (e-1) \cdot \frac{e^2-1}{2} = \frac{(e-1)(e^2-1)}{4} = \frac{(e-1)^2(e+1)}{4}.$$

(c) $\iiint_E y \, dV$, $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq x + y + 5$.

$$= \int_0^1 \int_0^1 y(x+y+5) \, dy \, dx = \int_0^1 \left[\frac{xy^2}{2} + \frac{y^3}{3} + \frac{5y^2}{2} \right]_0^1 \, dx = \int_0^1 \left(\frac{x}{2} + \frac{1}{3} + \frac{5}{2} \right) \, dx = \left[\frac{x^2}{4} + \frac{17x}{6} \right]_0^1 = \frac{1}{4} + \frac{17}{6}.$$

Exercise 13.2. (a) $\iiint_E z \, dV$ for tetrahedron; (b) volume between two paraboloids.

(a) Tetrahedron: $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$, $0 \leq z \leq 1 - x - y$.

$$\begin{aligned} \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z \, dz \, dy \, dx &= \int_0^1 \int_0^{1-x} \frac{(1-x-y)^2}{2} \, dy \, dx \\ &= \frac{1}{2} \int_0^1 \left[-\frac{(1-x-y)^3}{3} \right]_0^{1-x} dx = \frac{1}{6} \int_0^1 (1-x)^3 \, dx = \frac{1}{6} \left[-\frac{(1-x)^4}{4} \right]_0^1 = \frac{1}{24}. \end{aligned}$$

(b) Intersection of $z = x^2 + y^2$ and $z = 36 - 3(x^2 + y^2)$: $4(x^2 + y^2) = 36 \Rightarrow r = 3$.

$$V = \int_0^{2\pi} \int_0^3 [(36 - 3r^2) - r^2] r \, dr \, d\theta = 2\pi \int_0^3 (36r - 4r^3) \, dr = 2\pi [18r^2 - r^4]_0^3 = 2\pi(162 - 81) = \boxed{162\pi}.$$

14. Triple Integrals in Cylindrical Coordinates

Exercise 14.1. Use cylindrical coordinates to evaluate.

(a) $\iiint_E (x^2 + y^2) \, dV$, inside $r = 2$, $-3 \leq z \leq 5$.

$$= \int_0^{2\pi} \int_0^2 \int_{-3}^5 r^2 \cdot r \, dz \, dr \, d\theta = 2\pi \cdot 8 \int_0^2 r^3 \, dr = 16\pi \left[\frac{r^4}{4} \right]_0^2 = 16\pi \cdot 4 = \boxed{64\pi}.$$

(b) $\iiint_E \sqrt{x^2 + y^2} \, dV$, below $z = 4$, above $z = r^2$.

$$= \int_0^{2\pi} \int_0^2 \int_{r^2}^4 r \cdot r \, dz \, dr \, d\theta = 2\pi \int_0^2 r^2(4 - r^2) \, dr = 2\pi \int_0^2 (4r^2 - r^4) \, dr = 2\pi \left[\frac{4r^3}{3} - \frac{r^5}{5} \right]_0^2 = 2\pi \left(\frac{32}{3} - \frac{32}{5} \right)$$

(c) Convert: $r \in [0, 2]$, $\theta \in [0, 2\pi]$, $z \in [r, 2]$.

$$= \int_0^{2\pi} \int_0^2 \int_r^2 r^2 \cdot r \, dz \, dr \, d\theta = 2\pi \int_0^2 r^3(2 - r) \, dr = 2\pi \int_0^2 (2r^3 - r^4) \, dr = 2\pi \left[\frac{r^4}{2} - \frac{r^5}{5} \right]_0^2 = 2\pi \left(8 - \frac{32}{5} \right) =$$

Exercise 14.2. Volume of upper hemisphere of radius a using cylindrical coordinates.

$$V = \int_0^{2\pi} \int_0^a \int_0^{\sqrt{a^2 - r^2}} r \, dz \, dr \, d\theta = 2\pi \int_0^a r \sqrt{a^2 - r^2} \, dr.$$

Let $u = a^2 - r^2$, $du = -2r \, dr$:

$$= 2\pi \cdot \frac{1}{2} \int_0^{a^2} \sqrt{u} \, du = \pi \left[\frac{2u^{3/2}}{3} \right]_0^{a^2} = \frac{2\pi a^3}{3}.$$

15. Triple Integrals in Spherical Coordinates

Exercise 15.1. Use spherical coordinates to evaluate.

(a) $\iiint_B e^{(\rho^2)^{3/2}} \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta$ with $\rho \in [0, 1]$.

$$= \int_0^{2\pi} d\theta \int_0^\pi \sin \varphi \, d\varphi \int_0^1 e^{\rho^3} \rho^2 \, d\rho = 2\pi \cdot 2 \cdot \left[\frac{e^{\rho^3}}{3} \right]_0^1 = 4\pi \cdot \frac{e-1}{3} = \frac{4\pi(e-1)}{3}.$$

(b) $\iiint_H (x^2 + y^2 + z^2)^2 \, dV$, upper hemisphere $\rho \leq 2$, $0 \leq \varphi \leq \pi/2$.

$$= \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin \varphi \, d\varphi \int_0^2 \rho^4 \cdot \rho^2 \, d\rho = 2\pi \cdot [-\cos \varphi]_0^{\pi/2} \cdot \left[\frac{\rho^7}{7} \right]_0^2 = 2\pi \cdot 1 \cdot \frac{128}{7} = \frac{256\pi}{7}.$$

(c) $\iiint_E z \, dV$, above cone $\varphi = \pi/4$, inside sphere $\rho = 2$.

$$= \int_0^{2\pi} d\theta \int_0^{\pi/4} \int_0^2 \rho \cos \varphi \cdot \rho^2 \sin \varphi \, d\rho \, d\varphi = 2\pi \left[\frac{\rho^4}{4} \right]_0^2 \int_0^{\pi/4} \sin \varphi \cos \varphi \, d\varphi = 2\pi \cdot 4 \cdot \left[\frac{\sin^2 \varphi}{2} \right]_0^{\pi/4} = 8\pi \cdot \frac{1}{4} = 2\pi.$$

Exercise 15.2. (a) Volume inside sphere $\rho = 2$, above cone $\varphi = \pi/4$; (b) $\iiint_E (x^2 + y^2) \, dV$ for ball of radius 3.

(a) $\varphi \in [0, \pi/4]$, $\rho \in [0, 2]$:

$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^2 \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta = 2\pi \cdot \frac{8}{3} \cdot [-\cos \varphi]_0^{\pi/4} = \frac{16\pi}{3} \left(1 - \frac{1}{\sqrt{2}} \right) = \frac{8\pi(2 - \sqrt{2})}{3}.$$

(b) $x^2 + y^2 = \rho^2 \sin^2 \varphi$:

$$\iiint_E (x^2 + y^2) \, dV = \int_0^{2\pi} \int_0^\pi \int_0^3 \rho^2 \sin^2 \varphi \cdot \rho^2 \sin \varphi \, d\rho \, d\varphi \, d\theta = 2\pi \cdot \frac{3^5}{5} \cdot \int_0^\pi \sin^3 \varphi \, d\varphi.$$

$$\int_0^\pi \sin^3 \varphi \, d\varphi = \frac{4}{3}.$$

$$= 2\pi \cdot \frac{243}{5} \cdot \frac{4}{3} = \frac{648\pi}{5}.$$

16. Mixed Challenge Problems

Exercise 16.1. $T(x, y) = x^2 - xy + y^2$. (a) Critical points; (b) max/min on $x^2 + y^2 \leq 4$.

(a) $T_x = 2x - y = 0$, $T_y = -x + 2y = 0 \Rightarrow x = y = 0$. $D = T_{xx}T_{yy} - T_{xy}^2 = 2 \cdot 2 - (-1)^2 = 3 > 0$, $T_{xx} > 0$: global **minimum** at $(0, 0)$, $T = 0$.

(b) Interior: $T(0, 0) = 0$. Boundary $x = 2 \cos \theta$, $y = 2 \sin \theta$:

$$T = 4 \cos^2 \theta - 4 \cos \theta \sin \theta + 4 \sin^2 \theta = 4 - 2 \sin(2\theta).$$

Max when $\sin(2\theta) = -1$: $T = 6$ at $\theta = 3\pi/4$, i.e. $(-\sqrt{2}, \sqrt{2})$ (and $(\sqrt{2}, -\sqrt{2})$). Min when $\sin(2\theta) = 1$: $T = 2$ at $\theta = \pi/4$, i.e. $(\sqrt{2}, \sqrt{2})$.

Overall: **Absolute max** = 6, **absolute min** = 0.

Exercise 16.2. $f(x, y) = x^2 + 4xy + y^3$. (a) dy/dx if $f = 0$; (b) ∇f at $(1, -1)$; (c) directional derivative at $(1, -1)$ toward origin.

(a) By implicit differentiation:

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{2x + 4y}{4x + 3y^2}.$$

(b) $\nabla f = \langle 2x + 4y, 4x + 3y^2 \rangle$. At $(1, -1)$: $\nabla f = \langle 2 - 4, 4 + 3 \rangle = \langle -2, 7 \rangle$.

(c) Direction toward origin from $(1, -1)$: $\mathbf{d} = \langle -1, 1 \rangle$, $|\mathbf{d}| = \sqrt{2}$. Unit vector $\mathbf{u} = \langle -1, 1 \rangle / \sqrt{2}$.

$$D_{\mathbf{u}}f = \langle -2, 7 \rangle \cdot \frac{\langle -1, 1 \rangle}{\sqrt{2}} = \frac{2 + 7}{\sqrt{2}} = \frac{9\sqrt{2}}{2}.$$

Exercise 16.3. Evaluate $I = \int_{-\infty}^{\infty} e^{-x^2} dx$ via polar coordinates.

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \left[-\frac{e^{-r^2}}{2} \right]_0^{\infty} = 2\pi \cdot \frac{1}{2} = \pi.$$

Therefore $I = \sqrt{\pi}$. □

Exercise 16.4. Volume between paraboloid $z = x^2 + y^2$ and sphere $x^2 + y^2 + z^2 = 6$.

Choice of coordinates: cylindrical, since the paraboloid is $z = r^2$ and sphere is $r^2 + z^2 = 6$. Intersection: $r^2 + (r^2)^2 = 6 \Rightarrow r^4 + r^2 - 6 = 0 \Rightarrow (r^2 - 2)(r^2 + 3) = 0 \Rightarrow r = \sqrt{2}$, $z = 2$.

$$V = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{\sqrt{6-r^2}} r dz dr d\theta = 2\pi \int_0^{\sqrt{2}} r(\sqrt{6-r^2} - r^2) dr.$$

For $\int_0^{\sqrt{2}} r\sqrt{6-r^2} dr$: let $u = 6 - r^2$: $= \left[-\frac{(6-r^2)^{3/2}}{3} \right]_0^{\sqrt{2}} = -\frac{4^{3/2}}{3} + \frac{6^{3/2}}{3} = \frac{6\sqrt{6}-8}{3}$. For

$$\int_0^{\sqrt{2}} r^3 dr = \left[\frac{r^4}{4} \right]_0^{\sqrt{2}} = 1.$$

$$V = 2\pi \left(\frac{6\sqrt{6}-8}{3} - 1 \right) = \frac{2\pi(6\sqrt{6}-11)}{3}.$$

Exercise 16.5. Curve $F(x, y) = x \sin(y) - y \cos(x) = 0$. (a) Find dy/dx near $(0, 0)$; (b) tangent line at $(0, 0)$.

$$F_x = \sin(y) + y \sin(x), \quad F_y = x \cos(y) - \cos(x). \quad \text{At } (0, 0): \quad F_x = 0, \quad F_y = 0 - 1 = -1.$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{0}{-1} = 0.$$

(b) Slope = 0 at $(0, 0)$, and $F(0, 0) = 0 \cdot \sin 0 - 0 \cdot \cos 0 = 0$ (point is on the curve).
Tangent line: $y = 0$.

End of Solutions